

CO3 – AT2 – DESIGN TASK-MEDIUM

COURSE NAME : advanced wireless communication system
5G/6G communication systems

Course code : ECS5612

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"Design a Remote Patient Monitoring Communication Architecture"

CODE :

```
clc;

clear;

close all;

% Time (seconds)

time = 1:20;

% Simulated Patient Data

heart_rate = [72 73 74 75 76 78 80 79 77 76 75 74 73 72 71 72 73 74 75 76];

spo2 = [98 98 97 98 99 98 97 96 97 98 99 98 97 98 99 98 97 98 99 98];

temperature = [36.8 36.9 37.0 37.1 37.0 37.2 37.3 37.2 37.1 37.0 ...
               36.9 36.8 36.9 37.0 37.1 37.0 36.9 36.8 36.9 37.0];

% Plot Heart Rate

figure;

plot(time, heart_rate, '-o', 'LineWidth', 2);

grid on;

xlabel('Time (s)');

ylabel('Heart Rate (BPM)');

title('Remote Patient Monitoring - Heart Rate');

% Plot SpO2

figure;

plot(time, spo2, '-s', 'LineWidth', 2);
```

```

grid on;
xlabel('Time (s)');
ylabel('SpO2 (%)');
title('Remote Patient Monitoring - Oxygen Saturation');

% Plot Temperature
figure;
plot(time, temperature, '-^', 'LineWidth', 2);
grid on;
xlabel('Time (s)');
ylabel('Temperature (°C)');
title('Remote Patient Monitoring - Body Temperature');

% Alert Generation
if heart_rate(end) > 100 || heart_rate(end) < 60
    disp('ALERT: Abnormal Heart Rate Detected!');
else
    disp('Heart Rate is Normal.');
```

end

```

if spo2(end) < 95
    disp('ALERT: Low Oxygen Saturation!');
```

else

```

    disp('SpO2 is Normal.');
```

end


```

if temperature(end) > 37.5
    disp('ALERT: High Body Temperature!');
```

else

```

    disp('Body Temperature is Normal.');
```

end



